Effectiveness versus Detectability: Learned Physical-Layer Jamming against Learned Detectors

Bachelor's Thesis Project Proposal

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September 1, 2026

1 Introduction

Machine learning is now standard in physical-layer security. Convolutional classifiers on time–frequency representations of the received signal report jamming detection accuracies above 99 % on realistic data [1] and are increasingly treated as a solved building block, while the attacker side shifts from fixed heuristic jammers toward adversaries that learn their strategy from interaction [2].

A learned detector is, however, a model, and models can be attacked. Classifiers admit small targeted perturbations that flip their decisions [3] and transfer between models, so white-box access is not required [4], including for radio signal classifiers [5]. A jamming attack designed against a learned detector is exactly such an evasion attack, under two constraints the adversarial machine learning literature does not impose: the perturbation must be *physically realizable* through the attacker's own channel, and it must be *effective*, i.e. actually raise the error rate rather than merely change a label. The object of study is therefore a trade-off — how much degradation is achievable at a given detection probability. It is partly physical, since interference that flips bits necessarily perturbs the received constellation, but loose enough to leave design margin: interference that displaces symbols *across* their decision boundaries rather than along them causes the same errors at lower energy, and lower energy is harder to detect.

Two questions follow. First, learning only pays where there is structure to find. If the payload is independent and uniformly distributed — which scrambling is designed to ensure — the minimum-energy attack is available in closed form and a learner can at best rediscover it; any real advantage must come from structure that survives scrambling: deterministic protocol fields, the detector's decision surface, the channel, or the coordination problem arising when several attackers combine their transmissions. Second, both sides can adapt, and the interaction resembles a generative adversarial game [6] — but only offline, since a deployed detector has no ground-truth labels at run time and a deployed attacker cannot measure the error rate it induces. The practical question is then not who wins an unbounded arms race but what each round of adaptation *costs* — which existing work, including multi-agent formulations [10], has largely not quantified.

2 Research Questions

RQ1. What is the achievable trade-off between link degradation and detectability for physically realizable interference, and how does it depend on the noise level, the attacker's power budget, and the information it holds about the victim and the channel? The trade-off is characterized against a detector suite rather than one classifier, so that results describe the problem and not one trained model.

RQ2. Under which conditions does a learned attack strategy improve on a closed-form, energy-optimal one? This asks what must be present — deterministic protocol structure, exploitable channel information, a shapeable detector decision surface, or a coordination problem among several attackers — before learning separates from the closed form, and how large that separation is. A bounded or negative answer is admissible and informative.

RQ3. What does it cost either side to adapt once the other has moved? Treating the interaction as a round-based offline game — frozen detector, attacker optimized against it, detector retrained, attacker re-optimized — this asks how that cost is measured and how it behaves as the scenario becomes more realistic and the number of coordinating attackers grows.

3 Methodology

Minimal model first. The work starts from the smallest model in which the research questions remain meaningful and adds one layer at a time, each addition justified by a question the previous layer could not answer. The reason is not simplicity for its own sake: in a small model the defender's optimal decision rule is analytically available [7], which upgrades "this trained detector fails to see the attack" into the far stronger "no detector can do better than this". The base model is a single QPSK link over an additive white Gaussian noise channel attacked by one interferer under a hard power budget, with detection at the receiver from the received signal alone. The noise level is swept starting from the noiseless case: without noise the un-jammed constellation is a finite set of exact points and any perturbation is certainly detectable, so the stealth region exists only as a consequence of noise — making that explicit is part of the intended result.

Attackers and defenders. Attack strategies form a ladder of increasing knowledge, so every learned result is bracketed by closed-form references: no attacker (error floor and false-alarm rate); isotropic random-direction interference (no knowledge); minimum-energy interference displacing each received symbol just across its nearest decision boundary (closed form, no training — the real bar for any learner); full cancellation (omniscient, an upper bound rather than a realistic capability); and the learned attacker, which optimizes the error rate penalized by detection probability, with the power budget as a hard environment constraint rather than a reward term. The defender is likewise a suite: an energy detector at fixed false-alarm rate, a learned classifier following state-of-the-art spectrogram detectors [1], and — where the model admits it — the optimal likelihood-ratio test as reference.

Extensions, in order. (i) *Spatial / multi-agent*: several coordinating attackers whose transmissions superpose at the receiver, trained under centralized training with decentralized execution [8] through a multi-agent-native environment interface [9]. This is the intended destination of the thesis and the setting in which no closed-form solution exists; the training/execution asymmetry is a modelling constraint, since the error rate is available to a centralized critic during training whereas a deployed attacker observes only its own transmission, its own channel estimate and at most an acknowledgement from the victim. (ii) *Hardware realism*: carrier frequency, timing and phase offsets, modelling attackers that cannot synchronize perfectly — weakening the attacker and strengthening any result that survives. (iii) *Waveform realism*: a multicarrier waveform over a frequency-selective fading channel. This proposal fixes the progression, not its final form; the system model is restated as each extension is introduced.

Evaluation. Results are reported as bit and symbol error rate against detection probability at a fixed false-alarm rate. Two conventions are enforced: strategies are compared at *matched detectability* rather than at equal transmit power, since an attack that raises the error rate by raising its own signature is not an improvement; and every figure carries the full reference envelope from the no-attacker floor to the omniscient ceiling. Parameter studies cover noise level, power budget, the exploitable structure in the transmitted sequence (the direct test of RQ2), the number of attackers and legitimate users, and synchronization quality. For RQ3, adaptation cost is measured per round as the change in detector accuracy and false-alarm rate together with the data and training effort required, and the fraction of performance the attacker then recovers — a cost curve rather than a single winner. Experiments use Sionna [11] for the physical layer and PyTorch for the learned components.

Risks. The principal risk is that the learned attacker does not separate from the closed-form attack on a single link. This is anticipated rather than avoided: the structure ablation is scheduled early, before effort is invested in the learner, and a negative result redirects the work toward the coordinated setting, where no closed form exists. A second risk — results that look strong only because the attacker was granted unrealistic knowledge — is answered by the assumption ladder above, in which capabilities are removed in defined steps.

Work plan. **(1)** Model specification. **(2)** Minimal model: link, attacker ladder, detector suite, optimal-test reference. **(3)** Effectiveness–detectability characterization (RQ1). **(4)** Structure ablation and learned attacker at matched detectability (RQ2) — the decision point for whether the remaining emphasis falls on single-link design or on coordination. **(5)** Multi-agent extension (RQ2, RQ3). **(6)** Adaptation-cost rounds and synchronization realism (RQ3). **(7)** Multicarrier and fading channels; writing. Phases 1–3 are prerequisites; 5–7 are ordered by expected value and may be truncated from the end.

References

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